

Exp 16. To compare the performance of different classification algorithms.

Aim

To compare the performance of **Decision Tree** and **K-Nearest Neighbors (KNN)** classification algorithms using a dataset and evaluate their accuracy.

Software Required

- Python 3.x
- Scikit-learn Library
- NumPy
- Pandas (optional)

Algorithm

1. Import the required libraries.
2. Create the dataset and separate features and target variable.
3. Split the dataset into training and testing sets.
4. Train the Decision Tree classifier.
5. Train the KNN classifier.
6. Predict the test data using both models.
7. Calculate the accuracy of each model.
8. Compare the accuracies.
9. Predict the class of a new sample.

Dataset 1: Student Placement Prediction

CGPA Aptitude Score Communication Score Placement

9.2	90	88	Placed
8.8	85	82	Placed
8.5	80	79	Placed
7.2	72	70	Not Placed
6.8	68	65	Not Placed
9.0	92	90	Placed
7.5	75	74	Not Placed
8.6	84	81	Placed

CGPA Aptitude Score Communication Score Placement

6.5	60	58	Not Placed
8.9	88	86	Placed

Target Variable

Placement: Placed / Not Placed

New Sample

CGPA Aptitude Score Communication Score

8.7 86 84

Program

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

# Dataset
X = [
    [9.2, 90, 88],
    [8.8, 85, 82],
    [8.5, 80, 79],
    [7.2, 72, 70],
    [6.8, 68, 65],
    [9.0, 92, 90],
    [7.5, 75, 74],
    [8.6, 84, 81],
    [6.5, 60, 58],
    [8.9, 88, 86]
]

y = ["Placed", "Placed", "Placed", "Not Placed", "Not Placed",
     "Placed", "Not Placed", "Placed", "Not Placed", "Placed"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42)

dt = DecisionTreeClassifier()
dt.fit(X_train, y_train)
dt_acc = accuracy_score(y_test, dt.predict(X_test))

knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train, y_train)
```

Result

```
(X_train, X_test, y_train, y_test)
new_sample = [[8.7, 86, 84]]
print("Decision Tree Prediction:", dt.predict(new_sample)[0])
print("KNN Prediction:", knn.predict(new_sample)[0])
```

```
... Decision Tree Accuracy: 1.0
   KNN Accuracy: 1.0
   Decision Tree Prediction: Placed
   KNN Prediction: Placed
```

Dataset 1: Student Placement Prediction

CGPA Aptitude Score Communication Score Placement

9.2	90	88	Placed
8.8	85	82	Placed
8.5	80	79	Placed
7.2	72	70	Not Placed
6.8	68	65	Not Placed
9.0	92	90	Placed
7.5	75	74	Not Placed
8.6	84	81	Placed
6.5	60	58	Not Placed
8.9	88	86	Placed

Target Variable

Placement: Placed / Not Placed

New Sample

CGPA Aptitude Score Communication Score

8.7	86	84
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Program:

```

from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

X = [
    [8, 780, 5],
    [7, 760, 4],
    [6, 730, 5],
    [4, 620, 8],
    [3, 590, 9],
    [9, 810, 4],
    [5, 650, 7],
    [8, 770, 5],
    [4, 610, 8],
    [7, 750, 6]
]

y = ["Approved", "Approved", "Approved", "Rejected", "Rejected",
     "Approved", "Rejected", "Approved", "Rejected", "Approved"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42)

dt = DecisionTreeClassifier()
dt.fit(X_train, y_train)
dt_acc = accuracy_score(y_test, dt.predict(X_test))

knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train, y_train)

```

Output

```
print("KNN Prediction:", knn.predict(new_sample)[0])
```

```

... Decision Tree Accuracy: 1.0
    KNN Accuracy: 1.0
    Decision Tree Prediction: Placed
    KNN Prediction: Placed

```

Dataset 3: Disease Diagnosis

Temperature (°C)	Heart Rate (bpm)	Oxygen Level (%)	Disease
39.0	110	94	Positive
38.5	108	95	Positive
37.0	78	99	Negative
39.2	115	93	Positive
36.8	75	98	Negative
38.8	112	94	Positive
37.2	80	98	Negative
39.1	114	92	Positive
36.7	74	99	Negative
38.9	111	93	Positive

Target Variable

Disease: Positive / Negative

New Sample

Temperature	Heart Rate	Oxygen Level
38.7	109	94

Program:

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```
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

X = [
    [8, 780, 5],
    [7, 760, 4],
    [6, 730, 5],
    [4, 620, 8],
    [3, 590, 9],
    [9, 810, 4],
    [5, 650, 7],
    [8, 770, 5],
    [4, 610, 8],
    [7, 750, 6]
]

y = ["Approved", "Approved", "Approved", "Rejected", "Rejected",
     "Approved", "Rejected", "Approved", "Rejected", "Approved"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42)

dt = DecisionTreeClassifier()
dt.fit(X_train, y_train)
dt_acc = accuracy_score(y_test, dt.predict(X_test))

knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train, y_train)
knn_acc = accuracy_score(y_test, knn.predict(X_test))
```

Output:

```
print( KNN Prediction: , knn.predict(new_sample)[0])
```

```
Decision Tree Accuracy: 1.0
KNN Accuracy: 1.0
Decision Tree Prediction: Approved
KNN Prediction: Approved
```

Dataset 4: Employee Promotion Prediction

Experience (Years)	Performance Score	Training Hours	Promotion
10	95	50	Promoted
8	90	45	Promoted
7	88	40	Promoted
3	65	20	Not Promoted
2	60	18	Not Promoted
9	93	48	Promoted
4	70	25	Not Promoted
8	89	42	Promoted
2	58	15	Not Promoted
6	85	38	Promoted

Target Variable

Promotion: Promoted / Not Promoted

New Sample

Experience	Performance Score	Training Hours
7	90	44

Program:

```

from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

X = [
    [39.0, 110, 94],
    [38.5, 108, 95],
    [37.0, 78, 99],
    [39.2, 115, 93],
    [36.8, 75, 98],
    [38.8, 112, 94],
    [37.2, 80, 98],
    [39.1, 114, 92],
    [36.7, 74, 99],
    [38.9, 111, 93]
]

y = ["Positive", "Positive", "Negative", "Positive", "Negative",
     "Positive", "Negative", "Positive", "Negative", "Positive"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42)

dt = DecisionTreeClassifier()
dt.fit(X_train, y_train)
dt_acc = accuracy_score(y_test, dt.predict(X_test))

knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train, y_train)

```

Output:

```

... Decision Tree Accuracy: 1.0
   KNN Accuracy: 1.0
   Decision Tree Prediction: Positive
   KNN Prediction: Positive

```


Dataset 5: Vehicle Classification

Engine Capacity (cc) Weight (kg) Mileage (km/L) Vehicle Type

100	95	70	Bike
125	110	65	Bike
150	125	55	Bike
800	850	22	Car
1000	950	20	Car
1200	1050	18	Car
110	100	68	Bike
900	900	21	Car
140	120	58	Bike
1300	1100	17	Car

Target Variable

Vehicle Type: Bike / Car

New Sample

Engine Capacity Weight Mileage

135 118 60

Program:

[5]
✓ Os

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

X = [
    [10, 95, 50],
    [8, 90, 45],
    [7, 88, 40],
    [3, 65, 20],
    [2, 60, 18],
    [9, 93, 48],
    [4, 70, 25],
    [8, 89, 42],
    [2, 58, 15],
    [6, 85, 38]
]

y = ["Promoted", "Promoted", "Promoted", "Not Promoted",
     "Not Promoted", "Promoted", "Not Promoted",
     "Promoted", "Not Promoted", "Promoted"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42)

dt = DecisionTreeClassifier()
dt.fit(X_train, y_train)
dt_acc = accuracy_score(y_test, dt.predict(X_test))

knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train, y_train)
knn_acc = accuracy_score(y_test, knn.predict(X_test))
```

Output:

```
Decision Tree Accuracy: 1.0
KNN Accuracy: 1.0
Decision Tree Prediction: Bike
KNN Prediction: Bike
```